Necessity Entrepreneurship: Self-Employment*

David Chivers[†] Zhigang Feng[‡] Elisa Keller[§] Bo Li[¶]

March 9, 2026

Abstract

TO BE COMPLETED.

JEL Classification: E23, I10, O40.

Keywords:Occupational Choice, Entrepreneur, Misallocation,

*We thank

[†]Department of Economics, Durham Business School, DH1 3LB, david.chivers@durham.ac.uk

[‡]Department of Economics, University of Nebraska, Omaha, NE 68106, United States, z.feng2@gmail.com.

[§]Department of Economics, University of Exeter, elisa.keller@exeter.ac.uk

[¶]Department of Economics, Peking University, lb.harbin@gmail.com .

1 Introduction

Entrepreneurship is often associated with innovation, firm growth, and rising wages. But a large share of observed entrepreneurial activity is much smaller in scale: many entrepreneurs operate as self-employed nonemployers rather than as employer firms. That distinction matters for aggregate efficiency. If households enter business ownership mainly because viable wage employment is unavailable, a rise in entrepreneurship may reflect weak outside options rather than productive entrepreneurial dynamism. This paper studies that distinction formally. We define opportunity entrepreneurs as households that choose entrepreneurship even when a viable employment relationship is available, and necessity entrepreneurs as households that choose entrepreneurship only when that employment option is absent.

Empirical work usually approaches this distinction in two ways. One uses stated motives. In weighted U.S. GEM APS microdata for 2019--2021, between 41 and 50 percent of early-stage entrepreneurs report starting a business because jobs are scarce, while between 65 and 73 percent also report motives tied to building wealth or very high income. Those responses are informative, but they do not deliver a clean classification: necessity and opportunity signals can overlap within the same entrepreneurial spell. A second approach classifies entrepreneurs by prior labor-market status. That is also useful, but observed status at entry is not the counterfactual object we want. Workers may start a business while still employed because they anticipate worse job prospects, while some more growth-oriented entrepreneurs may pass through temporary non-employment before starting a firm.

Our approach takes those empirical margins seriously, but treats them as proxies rather than definitions. We instead use a structural, state-contingent definition based on occupational choice. In the self-employment framework developed below, an entrepreneur can operate either as a self-employed nonemployer or as an employer entrepreneur. That lets the model speak not only to entry, but also to the composition and scale of entrepreneurial activity.

The distinction is especially relevant when labor-market conditions deteriorate. In downturns, weaker outside options can push households toward low-scale business ownership even if employer-oriented entrepreneurship remains weak. Figure 1 illustrates that pattern in U.S. business counts by employment size. During the Great Recession, zero-employee businesses were comparatively resilient, while establishments with employees fell back. The figure is not best read as evidence of a broad-based entrepreneurship boom. Rather, it points to a shift in the composition of business activity toward the nonemployer margin.

Figure 1: U.S. business counts by employment size, indexed to 1997.

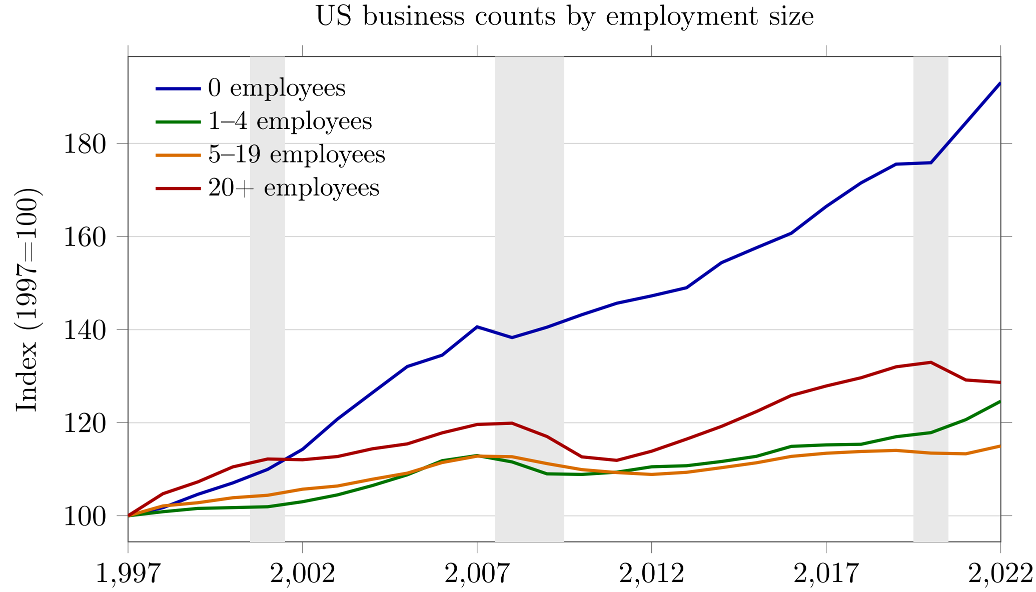
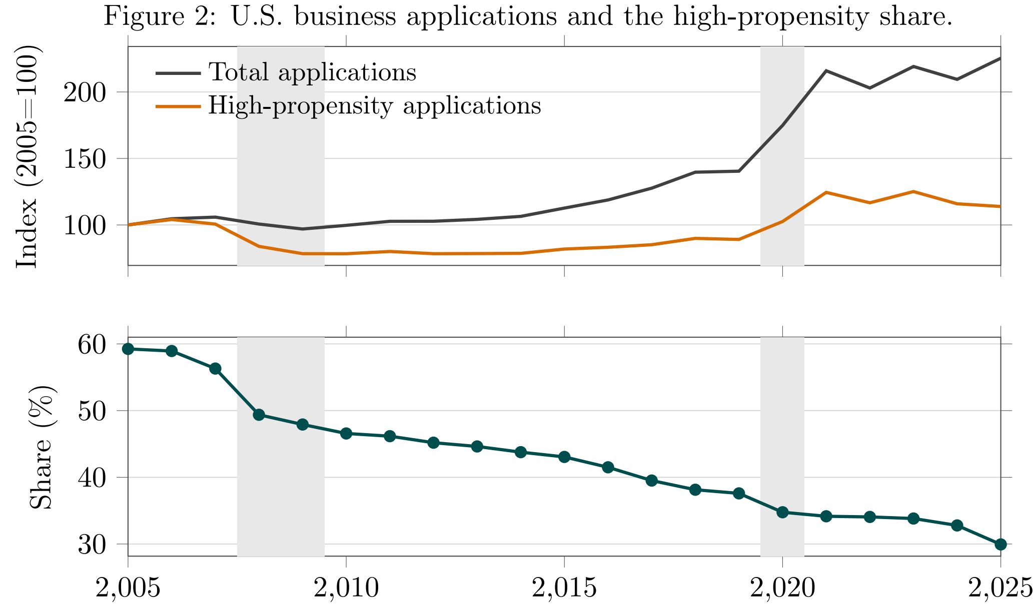


Figure 2 makes the same point on the flow margin. Using Census Business Formation Statistics, overall business applications dip over the recession window, but high-propensity applications fall much more sharply and their share of total applications declines. Recession conditions therefore do not simply raise or lower business creation in the aggregate. They tilt new business formation away from the employer-oriented margin. Together, the two figures motivate the paper's main measurement point: not all entrepreneurship is economically the same object.

Figure 2: U.S. business applications and the high-propensity share.



The paper makes three contributions. First, it provides a structural definition of necessity entrepreneurship that maps directly into state-contingent occupational choice. Second, it embeds that definition in a self-employment framework with heterogeneous education, entrepreneurial ability, closure risk, search frictions, and social insurance, allowing the model to speak jointly to entry, employer thresholds, and firm scale. Third, it interprets necessity entrepreneurship as a compositional margin: weak outside options can raise low-scale business ownership without strengthening employer entrepreneurship.

The rest of the paper first reviews the related literature, then presents the model and equilibrium, discusses the calibration, reports the baseline quantitative results, and finally studies the policy counterfactuals.

2 Literature Review

This paper connects three strands in a single quantitative framework.

First, we build on work that separates opportunity and necessity entrepreneurship in the data. Fairlie and Fossen (2019) propose an operational distinction based on pre-entry labor market status and show that opportunity and necessity entrepreneurship have different cyclical patterns. Sedlacek and Sterk (2017) show that startup cohorts differ in growth potential across the business cycle. This maps directly to our decomposition of entrepreneurial entry by prior employment state.

Second, we connect to the entrepreneurship-and-frictions literature. Buera (2009) shows in a dynamic model that borrowing constraints can materially change both entry and firm scale, with large welfare effects through undercapitalized businesses. Hurst and Pugsley (2011) documents that many small businesses are not primarily innovation-driven and that nonpecuniary motives are quantitatively important. This evidence supports modeling entrepreneurship as a broad occupational margin rather than only a high-innovation margin.

Third, our labor-market block follows the search-and-matching tradition. Petrongolo and Pissarides (2001) provides the aggregate matching-function synthesis that motivates our use of separation and job-finding transitions to discipline occupational flows.

Relative to these strands, our contribution is to embed an explicit opportunity-versus-necessity distinction in a heterogeneous-agent model with search frictions, incomplete insurance, and policy counterfactuals. This structure lets us quantify how unemployment risk and social insurance jointly shift entrepreneurial entry, firm scale, and composition.

3 The Model: Economic Environment

Consider a Lucas (1978) span of control model, where individuals differ in the level of human capital, h. As in Gomez and Kuhen (2017), individuals are educated to either the primary (P), secondary (S) or tertiary level (T). The level of education means that a tertiary educated individuals have a higher draw of human capital, on average, than secondary educated individuals, who in turn have a higher draw on average than primary educated individuals. This human capital draw is positively correlated with managerial ability x as well as labor productivity z.

At the start of the period agents enter as either unemployed ($i_w = 0$), employed ($i_w = 1$), or an entrepreneur ($i_w = 2$). Individuals face a choice of whether to become an entrepreneur, to continue working or to search for work. However, the value of employment is different depending on whether an agent enters the period as employed ($i_w = 1$) or not. Although there is a risk that an employee may become separated from a firm with probability ρ, this is less risky than searching for work which has a $p(\theta)$ probability of obtaining employment. Consequently, the expected value of employment is higher for those already in employment than those who are not. Additionally, the probability of separation is lower for individuals with higher levels of human capital and they are able to command a higher wage due to higher levels of productivity. So the value of employment, holding other things equal, is higher for those with higher human capital as well.

The innovation of this model is that the entrepreneurial decision results in two types of entrepreneurs emerge, opportunity-based entrepreneurs and necessity-based entrepreneurs. We define opportunity-based entrepreneurs are those whose value of entrepreneurship is higher, even if they had employment. Necessity-based entrepreneurs are those who would not have become entrepreneurs, if they had employment.

Timing: The timing of events can be summarized below. Occupational choice happens at the start of time t. Those choosing to become entrepreneurs run projects and choose capital and hired labor each period. If hired labor satisfies $n_h = 0$ the business is nonemployer self-employment, while $n_h > 0$ corresponds to employer entrepreneurship. Entrepreneurial businesses face closure risk $\rho_e(h)$. In the benchmark, closure moves the household out of business and into non-employment without automatic regular unemployment insurance eligibility. Those who were previously employed within a project can decide to carry on working but carry a risk of separation with probability ρ. If separation occurs these workers will claim unemployed benefits and be unable to look for work until the following period. After separation occurs, vacancies are posted and who were unemployed in the previous period are matched with firms. Those that are successful and match with with probability $p(\theta)$ go on to earn a wage. Whereas those that are unsuccessful will claim unemployment benefits and be unable to search again until the following period.

Unlike the search and matching literature there is no wage bargaining between workers and the firm at the end of matching. Instead the wage pins down the occupational choice decision which determines those wanting to become entrepreneurs and those who will search for work. Crucially, the worker only finds out whether they enter production and earn a wage after their occupational choice decision.

Figure 3: Timing for Matched Workers

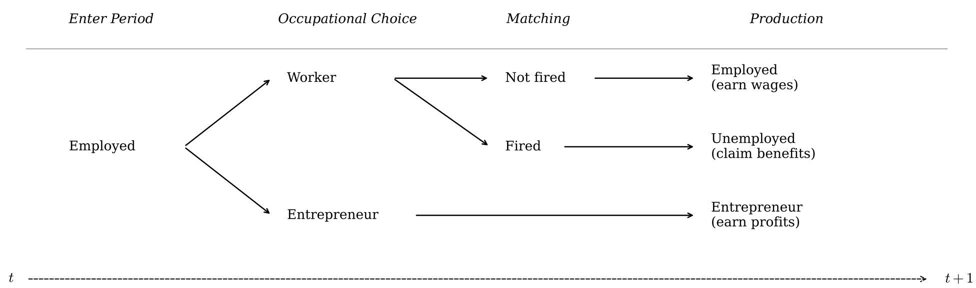
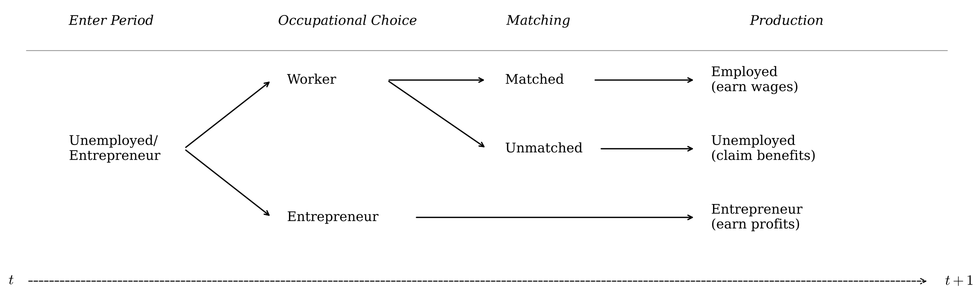


Figure 4: Timing for Unmatched Workers



3.1 Preferences, endowments and technology

Preferences: Consumption by an agent in period t is c_t, with utility given by $U(c_t)$.

Endowments: Agents are endowed with an initial capital asset, a_0, which can be used as an input in production. Each individual is endowed with human capital shock h^i. Human capital h is positively correlated with managerial talent x, labor productivity z and separation rate ρ.

Production: Firms use labor n and capital k to produce a single consumption good, y. Capital depreciates at a constant rate of δ.Managers can operate only one project.The functional form of the production function is:

$$y = xk^{\alpha}\left(\ell_e + \xi z n_h\right)^{\gamma} \quad \text{where } \alpha, \gamma > 0. \tag{1}$$

so self-employment is feasible at $n_h = 0$, while employer entrepreneurship corresponds to $n_h > 0$. Managerial talent x is determined by the Pareto distribution $F(x) = \left[1 + \frac{x-\varsigma}{\xi}\right]^{-\vartheta_i}$ where ϑ is the shape parameter, ς the location parameter and ξ is the scale parameter.

Factor remuneration: Firms rent capital at the common market rate r.

Government: The government runs a balanced budget each period and provides unemployment insurance which is financed through lump sum taxation τ, which is shared by the employee $(1-\psi)\tau$ and employer $\psi\tau$. This program guarantees each household an unemployment benefit of b.

3.2 The Firm's problem

The firm's problem is:

$$\max_{n,k} Xk^{\alpha}n^{\gamma} - (1+\tau\psi)wzn - rk - \kappa\frac{\rho zn}{q(\theta_t)} \tag{2}$$

employees must pay a share of unemployment insurance τ to all workers who produce output. Firms have to pay search costs κ for all vacancies posted. The number of vacancies posted is designed to replace any lost workers, ρzn. This, however, is affected by the firm-hiring rate $q(\theta_t) = \bar{m}\left(\frac{v_t}{u_t}\right)^{-\mu}$, where aggregate vacancies v_t is simply the aggregate level of all those who have separate from firms $\sum \rho zn$ within the period and u_t is the total unemployed looking for work within that period (see appendix).

When the market is tight and the hiring rate is low, firms have to search more and thus face higher search costs in order to replace lost workers. This is because $0 < q(\theta_t) < 1$,so a lower $q(\theta_t)$ increases $\frac{\rho zn}{q(\theta_t)}$. If the firm lose 10 workers, they want to hire 10 workers to replace them. But if the hiring rate, $q(\theta_t)$, is 50% then they will face double the amount of search costs than if $q(\theta_t)$ were 100%. The upshot of this is that when markets are tight, the size of the desired firm size n decreases ex ante , as the cost of replacing workers is high and so entrepreneurs would prefer to run smaller firms.

To avoid issues of directed search, we assume that employees are randomly matched with firms (as in Chivers et al., 2015) and each firm employees a distribution of workers across the human capital spectrum such that each firm employs the market average.

4 Optimal behavior and equilibrium

In any time period t agents are distinguished by (h, a, x, i, z). The timing of the economy is given as follows.

1. Households enter each new period with assets a and an employment status i_w

2. Idiosyncratic shocks h is drawn which determines x (managerial ability) and z (labor productivity), ρ seperation rate conditional on education level, P, S or T.

3. Households make an occupation decision: entrepreneur ($\mathcal{I}_e = 1$) or worker ($\mathcal{I}_e = 0$).

4. Capital and labor markets clear. A fraction ρ of employees separate from firms and $p(\theta)$ are hired to replace them

5. Households choose: consumption (c), borrowing/saving (a')

4.1 Firm manager

Firms are distinguished by their productivity realization x. In the self-employment version, entrepreneurs choose capital and hired labor while also supplying owner labor to the business. Let n_h denote hired labor and ℓ_e denote owner labor. When $n_h = 0$ the entrepreneur is self-employed; when $n_h > 0$ the entrepreneur is an employer. The static problem for a manager with talent x^i is

$$\max_{k,n_h \geq 0} xk^{\alpha}\left(\ell_e + \xi z n_h\right)^{\gamma} - zw(1+\psi\tau)n_h - rk - \kappa\frac{\rho z n_h}{q(\theta)} - f_{hire}\mathbf{1}\{n_h > 0\} \tag{3}$$

Self-employment is therefore the corner $n_h = 0$, while employer entrepreneurship corresponds to $n_h > 0$. For an interior employer choice, the first-order condition for hired labor is:

$$\gamma x k^{\alpha}\left(\ell_e + \xi z n_h\right)^{\gamma-1}\xi z = w(1+\psi\tau) + \frac{\rho\kappa}{q(\theta)} \tag{4}$$

$$n_h^*(k, x, \widetilde{w}) = \max\left\{0, \frac{1}{\xi z}\left[\left(\frac{\gamma x k^{\alpha}\xi z}{w(1+\psi\tau) + \frac{\rho\kappa}{q(\theta)}}\right)^{\frac{1}{1-\gamma}} - \ell_e\right]\right\} \tag{5}$$

Substituting (5) into (3) yields the entrepreneur's profit function:

$$\pi(k, n_h^*, x; \widetilde{w}) = xk^{\alpha}\left(\ell_e + \xi z n_h^*\right)^{\gamma} - z n_h^* w(1+\psi\tau) - rk - \kappa\frac{\rho z n_h^*}{q(\theta)} - f_{hire}\mathbf{1}\{n_h^* > 0\} \tag{6}$$

4.1.1 Capital

Initially we assume capital markets function perfectly, so the interior capital choice satisfies

$$\alpha x k^{\alpha-1}\left(\ell_e + \xi z n_h\right)^{\gamma} = r.$$

4.2 Entrepreneurs

Entrepreneurs maximize expected discounted utility of consumption

with $n_h = 0$ denoting nonemployer self-employment and $n_h > 0$ denoting employer entrepreneurship.

$$\max_{\{c_t, a_{t+1}, k_t, n_{h,t}\}} \mathbb{E}\sum_{t=0}^{\infty}\beta^t U(c_t)$$

subject to the following budget constraint:

$$c + a' \leq (1 + r - \delta)a + \pi(k, n_h, x; \tilde{r}, w, \tau) - \tau_\pi \tag{7}$$

$$a' \geq -\bar{a}$$

4.3 Employed

The workforce (previous periods employed who choose to reapply for work) maximize expected discounted utility of consumption

$$\max_{\{c_t, a_{t+1}\}} \mathbb{E} \sum_{t=0}^{\infty} \beta^t U(c_t)$$

subject to the following budget constraint:

$$c + a' \leq (1 + r - \delta)a + (1 - \rho)\left(wz(1 - (1 - \psi)\tau_y)\right) + \rho b \tag{8}$$

$$a' \geq -\bar{a}$$

where b is the unemployment benefit.

4.4 Unemployed

The unemployed maximize expected discounted utility of consumption

$$\max_{\{c_t, a_{t+1}\}} \mathbb{E} \sum_{t=0}^{\infty} \beta^t U(c_t)$$

subject to the following budget constraint:

$$c + a' \leq (1 + r - \delta)a + p(\theta)\left(wz(1 - (1 - \psi)\tau_y)\right) + (1 - p(\theta))b \tag{9}$$

$$a' \geq -\bar{a}$$

where b is the unemployment benefit.

4.5 Government

The government runs a balanced budget with a lump-sum taxes $T(y) = (y - \lambda_p y^{1-\tau_p})$

4.6 The household's problem

Let i_w denote the household's labor market status in the previous period, with $i_w = 2$ indicating being entrepreneur, $i_w = 1$ being employed and $i_w = 0$ being unemployed. Let $\mathcal{I}_e$ indicate occupational choice, where $\mathcal{I}_e = 1$ if the household is an entrepreneur and $\mathcal{I}_e = 0$ if the household is a worker.

Let $\mathbb{E}_{\rho(h)}$ denote the continuation operator for employed workers who face job separation, let $\mathbb{E}_{p(\theta)}$ denote the continuation operator for households searching for work, and let $\mathbb{E}_{\rho_e(h)}$ denote the continuation operator for entrepreneurs facing business closure. We write $\mathbf{V}_0$ for the continuation value after business closure; in the benchmark this state does not automatically receive regular unemployment insurance. More specifically,

$$\mathbb{E}_{p(\theta)}\mathbf{V}(a', h, x', i'_w) = p(\theta)\mathbf{V}(a', h, x', i'_w = 1) + (1 - p(\theta))\mathbf{V}(a', h, x', i'_w = 0)$$

$$\mathbb{E}_{\rho(h)}\mathbf{V}(a', h, x', i'_w) = (1 - \rho(h))\mathbf{V}(a', h, x', i'_w = 1) + \rho(h)\mathbf{V}(a', h, x', i'_w = 0)$$

$$\mathbb{E}_{\rho_e(h)}\mathbf{V}(a', h, x', i'_w) = (1 - \rho_e(h))\mathbf{V}(a', h, x', i'_w = 2) + \rho_e(h)\mathbf{V}_0(a', h, x', i'_w = 0)$$

$$\begin{aligned}
\mathbf{V}(a, h, x, i_w) &= \max\left\{V^w(a, h, x, i_w), V^e(a, h, x)\right\} \\
V^w(a, h, x, i_w) &= \max_{\{a', c\}}\left\{u(c) + \iota_{i_w=1}\beta\mathbb{E}_{\rho(h)}\mathbf{V}(a', h, x', i'_w) + \iota_{i_w\neq 1}\beta\mathbb{E}_{p(\theta)}\mathbf{V}(a', h, x', i'_w)\right\} \\
V^e(a, h, x) &= \max_{\{a', c, k, n_h \geq 0\}}\left\{u(c) + \chi_{se}\mathbf{1}\{n_h = 0\} + \chi_{emp}\mathbf{1}\{n_h > 0\} + \beta\mathbb{E}_{\rho_e(h)}\mathbf{V}(a', h, x', i'_w)\right\}
\end{aligned}$$

subject to

$$c + a' \leq (1 - \tilde{r} - \delta)a + inc \tag{10}$$

$$inc = \begin{cases} ra + \widetilde{w}_n & \text{if } \mathcal{I}_e = 0,\ i_w = 1 \\ ra + \widetilde{w}_u & \text{if } \mathcal{I}_e = 0,\ i_w \neq 1 \\ ra + \pi(k, n_h, x; \tilde{r}, w, \tau) & \text{if } \mathcal{I}_e = 1 \end{cases}$$

where inc is the earnings of the worker or entrepreneur, with entrepreneurial income coming from either self-employment ($n_h = 0$) or employer entrepreneurship ($n_h > 0$), $\widetilde{w}_n = (1 -$

$\rho)\left\{zw(1-(1-\psi)\tau)\right\}+\rho b$ and $\widetilde{w}_u = p(\theta)\left\{zw(1-(1-\psi)\tau)\right\}+\{1-p(\theta)\}b$.

4.7 Equilibrium

Let $\hat{h}$ be the critical value of human capital that determines occupational choice which is affected by both employment status and education level. If $h \geq \hat{h}$ then individuals choose entrepreneurship over becoming a worker. The critical value of human capital is always higher for those who have employment than those who are unmatched with a firm $\hat{h}_{i_w\neq 1} < \hat{h}_{i_w=1}$. We define necessity entrepreneurship as those who would not have become an entrepreneur *if* they had an employment contract:

$$\hat{h}_{i_w\neq 1} < h < \hat{h}_{i_w=1}.$$

We define opportunity-driven entrepreneurship as those who would become an entrepreneur *regardless* of circumstance:

$$h \geq \hat{h}_{i_w=1} \geq \hat{h}_{i_w\neq 1}$$

As there are three education levels, there are a total of six critical values of human capital: two for primary educated matched or unmatched, $\hat{h}^P_{i_w=1}$ and $\hat{h}^P_{i_w\neq 1}$ two for secondary educated, $\hat{h}^S_{i_w=1}$ and $\hat{h}^S_{i_w\neq 1}$, and two for tertiary educated, $\hat{h}^T_{i_w=1}$ and $\hat{h}^T_{i_w\neq 1}$.

5 Calibration

The benchmark self-employment calibration keeps the core preference, search, and production parameters from the baseline model and adds only the objects needed to separate own-account self-employment from employer entrepreneurship. Table 1 reports the benchmark values. The main additions are an education-specific entrepreneur closure risk, a partial capital-loss rule on closure, and a fixed employer-threshold cost paid when hired labor becomes strictly positive.

Table 1: Benchmark self-employment calibration

Parameter	Benchmark value	Discipline / interpretation
β	0.94	Discount factor inherited from the benchmark calibration.
σ	2.0	CRRA risk aversion.
$\bar{m}$	0.90	Matching-efficiency parameter in the worker labor market.
μ	0.68	Matching elasticity.
κ	0.035	Proportional vacancy and replacement cost in hiring.
x-share by education	$\{0.782, 0.802, 0.822\}$	Education-specific productivity-share terms inherited from the benchmark model.
$\rho(h)$	$\{0.0383, 0.0288, 0.0136\}$	Worker separation rates by education.
b	0.40	Benchmark unemployment-insurance replacement rate.
$\underline{c}$	0.30	Benchmark consumption floor.
$\rho_e(h)$	$\{0.168, 0.150, 0.144\}$	Entrepreneur closure risk by education; average level set at 0.15 and gradient disciplined by self-employment-to-unemployment hazards in Grashuis (2021).
λ_k	0.25	Fraction of installed entrepreneurial capital lost on closure.
ℓ_o	1.0	Owner labor in the self-employment technology, normalized to one.
x_{SE} scale	1.0	Productivity scale on the zero-hire self-employment corner, normalized in the benchmark.
f_{hire}	0.25	Fixed employer-threshold cost paid when hired labor becomes strictly positive; benchmark value chosen to match the self-employed versus employer composition, with robustness checks at 0.15 and 0.35.

For entrepreneur closure, we target an average annual exit rate of 0.15 and use Grashuis (2021) only to discipline the education gradient, yielding $\rho_e(h) = \{0.168, 0.150, 0.144\}$. Following regular U.S. unemployment-insurance rules, business closure removes the household from entrepreneurship but does not automatically grant regular UI eligibility. We pair that exit shock with a 25 percent loss of installed business capital, so closure affects both occupational status and next-period net worth.

The employer-threshold cost is set to $f_{hire} = 0.25$ in the benchmark. This cost is distinct from the proportional hiring cost governed by κ: κ captures vacancy and replacement costs that scale with hired labor, while f_{hire} captures the discrete hurdle between remaining self-employed and becoming an employer. We treat f_{hire} as a disciplined quantitative choice rather than as a directly estimated structural object, and Appendix robustness shows that the main composition results survive at nearby values of 0.15 and 0.35.

6 Baseline Results

The benchmark equilibrium delivers entrepreneurship of 17.8 percent of households. Within entrepreneurship, 57.7 percent of businesses are self-employed nonemployers and 42.3 percent are employer entrepreneurs. The composition differs sharply by education: low-education entrepreneurs are mostly self-employed, middle-education entrepreneurs are more balanced, and high-education entrepreneurs are predominantly employers. Table 2 also shows strong scale heterogeneity. Average entrepreneur labor demand is 6.16 overall, but ranges from 1.18 among low-education entrepreneurs to 22.68 among high-education entrepreneurs. Conditional on operating as employers, firm size remains substantial across all education groups, so the main benchmark margin is the composition shift between self-employment and employer entrepreneurship rather than the disappearance of employer firms.

Table 2: Benchmark self-employment equilibrium: baseline results

	Baseline
Entrepreneur share	
All households	0.1781
Low education	0.7187
Middle education	0.1329
High education	0.0725
Output per household	
All households	1288.2102
Low education	211.3357
Middle education	760.4220
High education	2656.0306
Average entrepreneur size n	
All entrepreneurs	6.1587
Low education	1.1767
Middle education	6.4030
High education	22.6811
Average employer size n	
Employer entrepreneurs, all education	14.5528
Low education	6.7829
Middle education	11.8076
High education	25.6474
Average entrepreneur capital k	
All entrepreneurs	98.1596
Low education	31.5415
Middle education	101.8159
High education	317.7547
Self-employed among entrepreneurs ($n_h = 0$)	
All education	0.5768
Low education	0.8265
Middle education	0.4577
High education	0.1157

7 Quantitative Analysis

This section evaluates the policy and benchmark counterfactuals around the self-employment calibration. The key object is not only the overall entrepreneurship rate, but the composition of entrepreneurship between nonemployer self-employment and employer entrepreneurship, together with the change in average entrepreneurial scale.

7.1 Unemployment insurance ladder

The first benchmark experiment lowers unemployment insurance while holding the benchmark self-employment calibration fixed. The main pattern is monotone: lower UI raises entrepreneurship, shifts entrepreneurs toward self-employment, and reduces average entrepreneurial scale. This is the central necessity-entrepreneurship result in the benchmark self-employment model.

Table 3: Benchmark self-employment equilibrium: unemployment insurance ladder

	Baseline (UI=0.40)	Low UI (UI=0.05)	No UI (UI=0.00)
Entrepreneur share			
All households	0.1781	0.2888	0.3425
Low education	0.7187	0.7964	0.8020
Middle education	0.1329	0.3054	0.3837
High education	0.0725	0.0791	0.1036
Average entrepreneur size n			
All entrepreneurs	6.1587	3.3198	2.5529
Low education	1.1767	0.8905	0.8258
Middle education	6.4030	2.5539	1.8831
High education	22.6811	17.4690	11.9016
Average employer size n			
Employer entrepreneurs, all education	14.5528	11.7813	10.9562
Low education	6.7829	6.7511	6.4060
Middle education	11.8076	8.8790	8.3019
High education	25.6474	22.6782	21.2851
Self-employed among entrepreneurs ($n_h = 0$)			
All education	0.5768	0.7182	0.7670
Low education	0.8265	0.8681	0.8711
Middle education	0.4577	0.7124	0.7732
High education	0.1157	0.2297	0.4408
Employer entrepreneurs among entrepreneurs ($n_h > 0$)			
All education	0.4232	0.2818	0.2330
Low education	0.1735	0.1319	0.1289
Middle education	0.5423	0.2876	0.2268
High education	0.8843	0.7703	0.5592

7.2 No unemployment risk

This counterfactual removes worker unemployment risk by fixing job finding at one and worker separation at zero, while holding the benchmark tax rate fixed. It is useful as an upper-bound labor-market benchmark, but it is not a full frictionless efficient allocation: entrepreneurial risk, borrowing limits, the employer-threshold cost, and occupational heterogeneity remain in place. For that reason, we interpret it as a worker-side wedge comparison rather than as a complete misallocation accounting exercise.

Table 4: Benchmark self-employment equilibrium: baseline versus no unemployment risk

	Baseline benchmark	No unemployment risk
Entrepreneur share		
All households	0.1781	0.1929
Low education	0.7187	0.4619
Middle education	0.1329	0.1732
High education	0.0725	0.1352
Output per household		
All households	1288.2102	1462.5170
Low education	211.3357	263.6092
Middle education	760.4220	853.0274
High education	2656.0306	3026.3699
Average entrepreneur size n		
All entrepreneurs	6.1587	5.8069
Low education	1.1767	1.5387
Middle education	6.4030	4.9294
High education	22.6811	13.0432
Average employer size n		
Employer entrepreneurs, all education	14.5528	9.6321
Low education	6.7829	6.7484
Middle education	11.8076	7.4950
High education	25.6474	14.1509
Self-employed among entrepreneurs ($n_h = 0$)		
All education	0.5768	0.3971
Low education	0.8265	0.7720
Middle education	0.4577	0.3423
High education	0.1157	0.0783
Employer entrepreneurs among entrepreneurs ($n_h > 0$)		
All education	0.4232	0.6029
Low education	0.1735	0.2280
Middle education	0.5423	0.6577
High education	0.8843	0.9217

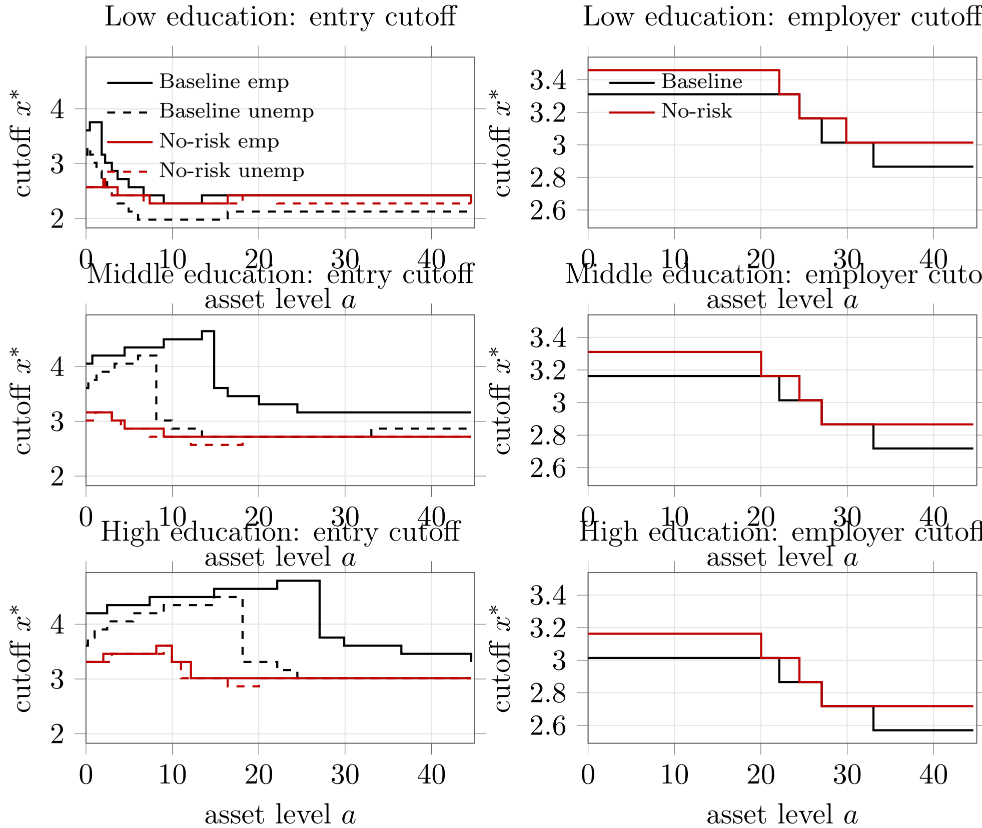


Figure 5: Benchmark self-employment cutoff comparisons using the current saved baseline and no-unemployment-risk policy outputs. Left panels plot the worker-to-entrepreneur occupational cutoff by prior labor-market status. Right panels plot the within-entrepreneur employer threshold implied by the static labor-demand policy; below that line entrepreneurship is self-employment, while above it entrepreneurship uses hired labor.

Relative to the benchmark, removing worker unemployment risk shifts the composition of entrepreneurship sharply toward employer firms. The self-employed share falls from 57.7 percent to 39.7 percent, while the employer share rises from 42.3 percent to 60.3 percent. Output per household also rises from 1288.2 to 1462.5. That makes the exercise useful for showing that the baseline self-employment mass is tightly linked to worker-side labor-market risk rather than only to the employer-threshold cost.

8 MIT Shock

The next step is a recession-style MIT shock built around a labor-market deterioration rather than a firm-side productivity collapse. As a first steady-state precursor, we increase low-education worker separation and compare the new stationary equilibrium with the benchmark. This is the right shock for the mechanism in the paper because it weakens worker outside options directly and can therefore raise broad entrepreneurship even while employer-oriented activity weakens.

Table 5: Benchmark self-employment equilibrium: downturn steady state with higher low-education separation

	Baseline benchmark	Downturn steady state
Entrepreneur share		
All households	0.1781	0.1868
Low education	0.7187	0.8314
Middle education	0.1329	0.1269
High education	0.0725	0.0725
Output per household		
All households	1288.2102	1260.7666
Low education	211.3357	156.1744
Middle education	760.4220	738.9283
High education	2656.0306	2627.1813
Average entrepreneur size n		
All entrepreneurs	6.1587	5.7981
Low education	1.1767	1.0113
Middle education	6.4030	6.6120
High education	22.6811	22.4394
Average employer size n		
Employer entrepreneurs, all education	14.5528	14.3264
Low education	6.7829	6.5555
Middle education	11.8076	11.6948
High education	25.6474	25.2752
Self-employed among entrepreneurs ($n_h = 0$)		
All education	0.5768	0.5953
Low education	0.8265	0.8457
Middle education	0.4577	0.4346
High education	0.1157	0.1122
Employer entrepreneurs among entrepreneurs ($n_h > 0$)		
All education	0.4232	0.4047
Low education	0.1735	0.1543
Middle education	0.5423	0.5654
High education	0.8843	0.8878

The downturn steady state has the intended sign pattern. Broad entrepreneurship rises slightly from 0.1781 to 0.1868, the self-employed share rises from 0.5768 to 0.5953, and average entrepreneur labor demand falls from 6.1587 to 5.7981. Output per household declines modestly from 1288.2 to 1260.8. This gives a disciplined starting point for the transition-path MIT-shock section that follows in the next draft.

9 Conclusion

This paper develops a high-level quantitative framework for studying necessity entrepreneurship as an equilibrium outcome of occupational choice under labor market frictions. By defining necessity and opportunity entrepreneurship through counterfactual choices, the model separates compositional shifts in entry from changes in aggregate entrepreneurship rates.

The main message is that unemployment risk and social insurance jointly shape who becomes an entrepreneur. Reducing insurance can raise entrepreneurship on paper, but part of this increase reflects necessity entry associated with lower average firm scale. Conversely, stronger insurance can reduce necessity entry and improve sorting, even when effects on total entrepreneurship are modest. These results support evaluating entrepreneurship policy through composition, not levels alone.

Several extensions remain important for the next draft. First, the MIT-shock section should be completed with a deterministic transition path built from the downturn steady-state comparison reported here. Second, the fixed-tax and hiring-cost robustness checks belong in the appendix once the remaining fixed-tax benchmark run is completed cleanly. Third, the role of wealth and borrowing constraints should be examined more directly, because the current asset-dependent scale block likely already embeds important wealth-to-scale effects even before an explicit credit-friction wedge is added.

Appendix

Robustness

The employer-threshold benchmark is disciplined locally around $f_{hire} = 0.25$ rather than being treated as a single lucky parameter choice. The appendix reports nearby values to show that the main composition result is robust within a tight bracket.

Table 6: Employer-threshold cost robustness in the benchmark baseline equilibrium

	$f_{\text{hire}} = 0.15$	$f_{\text{hire}} = 0.25$	$f_{\text{hire}} = 0.35$
Entrepreneur share (all households)	0.1786	0.1781	0.1767
Avg entrepreneur labor demand n	6.1274	6.1587	6.2336
Avg entrepreneur capital k	97.8587	98.1596	98.8487
Self-employed among entrepreneurs ($n_h = 0$)			
All education	0.5679	0.5768	0.5926
Low education	0.8190	0.8265	0.8494
Middle education	0.4482	0.4577	0.4637
High education	0.1109	0.1157	0.1215
Employer entrepreneurs among entrepreneurs ($n_h > 0$)			
All education	0.4321	0.4232	0.4074
Low education	0.1810	0.1735	0.1506
Middle education	0.5518	0.5423	0.5363
High education	0.8891	0.8843	0.8785

Table 7: Employer-threshold cost robustness in the benchmark no-UI equilibrium

	$f_{\text{hire}} = 0.15$	$f_{\text{hire}} = 0.25$	$f_{\text{hire}} = 0.35$
Entrepreneur share (all households)	0.3348	0.3425	0.3401
Avg entrepreneur labor demand n	2.6566	2.5529	2.6138
Avg entrepreneur capital k	55.6364	54.0490	54.7019
Self-employed among entrepreneurs ($n_h = 0$)			
All education	0.7321	0.7670	0.7710
Low education	0.8523	0.8711	0.8767
Middle education	0.7324	0.7732	0.7767
High education	0.4062	0.4408	0.4433
Employer entrepreneurs among entrepreneurs ($n_h > 0$)			
All education	0.2679	0.2330	0.2290
Low education	0.1477	0.1289	0.1233
Middle education	0.2676	0.2268	0.2233
High education	0.5938	0.5592	0.5567

The fixed-tax no-UI comparison is being kept as appendix material rather than as a main-text quantitative table, because the self-employment fixed-tax path is coded but still computationally expensive. Once that target run is completed cleanly, it can be added here as a fiscal-closure robustness check for the benchmark UI experiment.

Computation

Given the values for parameters, the discretized spaces Ω_a for a, Ω_z for z and Ω_x for x. Each agent draw a human capital level h (interpreted as educational attainment) at the beginning of the economy, which will be fixed henceforth. h will determine their labor productivity $z \in \Omega_z$, the separation rate ρ, and managerial ability $x \in \Omega_x$. Let $i_w = 0$ indicating being entrepreneur, $i_w = 1$ being employed and $i_w = 2$ being unemployed. The numerical algorithm works as follows.

1. Set a tolerance $\epsilon > 0$.

2. Guess $\Phi^0 = \left(r^0, w^0, \rho^0, \tau_y^0\right)$. Solve for optimal household behavior:

$$f : (\varphi; \Phi) \to \left(c, a', \mathcal{I}_e, n, k\right),$$

where $\varphi = \{h, z, x, a, i_w\}$. We will use the method of value function iteration as follows.

   (a) Guess value function $\mathbf{V}^0 \left(\varphi; \Phi^0\right)$ and policy functions $f^0 \left(\varphi; \Phi^0\right)$.

   (b) Calculate the cutoff values of x_u^* (for previously unemployed worker, or entrepreneur) and x_n^* (for previously employed worker) via the following conditions.

$$\widetilde{w}_u = \pi(x_u^*) = \max_{k, n_h \geq 0} \left\{ x_u^* k^\alpha \left(\ell_e + \xi z n_h\right)^\gamma - w(1 + \psi\tau) z n_h - rk - \frac{\kappa \rho z n_h}{q(\theta)} - f_{hire} \mathbf{1}\{n_h > 0\} \right\} \tag{11}$$

$$\widetilde{w}_n = \pi(x_n^*) = \max_{k, n_h \geq 0} \left\{ x_n^* k^\alpha \left(\ell_e + \xi z n_h\right)^\gamma - w(1 + \psi\tau) z n_h - rk - \frac{\kappa \rho z n_h}{q(\theta)} - f_{hire} \mathbf{1}\{n_h > 0\} \right\} \tag{12}$$

where $\widetilde{w}_n$ and $\widetilde{w}_u$ denote the expected compensation for the employed worker and unemployed worker respectively. More specifically,

$$\widetilde{w}_n = (1 - \rho) \left\{ z w (1 - (1 - \psi)\tau) \right\} + \rho b \tag{13}$$

$$\widetilde{w}_u = p(\theta) \left\{ z w (1 - (1 - \psi)\tau) \right\} p(\theta)) b. \tag{14}$$

Noting that the separation rate ρ is pre-determined by the individual's initial human capital level h.

(c) Update value and policy functions:

$$\mathbf{V}^1(h,z,x,a,i_w) = \tag{15}$$
$$\max_{\{a',c,\mathcal{I}_e\}} \left\{ \begin{array}{c} \iota_{i_w=0}\cdot\left[\begin{array}{c}(1-\mathcal{I}_e)\cdot\mathbb{E}_{p(\theta)}\left(V_w+\beta\mathbb{E}\mathbf{V}^0(h,z',x',a',i_w')\right)\\ +\mathcal{I}_e\cdot\left(V_e+\beta\mathbb{E}\mathbf{V}^0(h,z',x',a',i_w'=2)\right)\end{array}\right]+ \\ \iota_{i_w=1}\cdot\left[\begin{array}{c}(1-\mathcal{I}_e)\cdot\mathbb{E}_{\rho}\left(V_w+\beta\mathbb{E}\mathbf{V}^0(h,z',x',a',i_w')\right)\\ +\mathcal{I}_e\cdot\left(V_e+\beta\mathbb{E}\mathbf{V}^0(h,z',x',a',i_w'=2)\right)\end{array}\right]+ \\ \iota_{i_w=2}\cdot\left[\begin{array}{c}(1-\mathcal{I}_e)\cdot\mathbb{E}_{p(\theta)}\left(V_w+\beta\mathbb{E}\mathbf{V}^0(h,z',x',a',i_w')\right)\\ +\mathcal{I}_e\cdot\left(V_e+\beta\mathbb{E}\mathbf{V}^0(h,z',x',a',i_w'=2)\right)\end{array}\right] \end{array} \right\}$$
$$f^1\left(\varphi;\Phi^0\right) = \arg\max \mathbf{V}^1\left(\varphi;\Phi^0\right)$$

where $\mathbb{E}_{p(\theta)}\left(V_w+\beta\mathbb{E}\mathbf{V}(a',x',i_w')\right)$ denotes the expected payoff for the individual who is previously unemployed worker, or entrepreneur, and decides to look for paid job, while $\mathbb{E}_{\rho}\left(V_w+\beta\mathbb{E}\mathbf{V}(a',x',i_w')\right)$ represents the expected payoff for the previously employed worker who decide to stay in workplace.

$$\begin{aligned} \mathbb{E}_{p(\theta)}\left(V_w+\beta\mathbb{E}\mathbf{V}(h,z',x',a',i_w')\right) &= p(\theta)\left(V_n+\beta\mathbb{E}\mathbf{V}(h,z',x',a',i_w'=1)\right) \qquad (16)\\ &\quad + (1-p(\theta))\left(V_u+\beta\mathbb{E}\mathbf{V}(h,z',x',a',i_w'=0)\right)\\ \mathbb{E}_{\rho}\left(V_w+\beta\mathbb{E}\mathbf{V}(h,z',x',a',i_w')\right) &= (1-\rho)\left(V_n+\beta\mathbb{E}\mathbf{V}(h,z',x',a',i_w'=1)\right) \qquad (17)\\ &\quad + \rho\left(V_u+\beta\mathbb{E}\mathbf{V}(h,z',x',a',i_w'=0)\right) \end{aligned}$$

To simplify the analysis, we assume that the individual who was previously employed and newly separated from the firm is not allowed to search for job at the current period. The measure for the individuals who are looking for jobs, u_t, the total vacancies v_t, and job matching rate $\theta_{n,t}$ are given as follows.

$$u_t = \left(\int_x \iota_{i_w=0} + \int_x \iota_{i_w=2}\right)\cdot\int_0^{x_u^*} \tag{18}$$
$$v_t = \int_x \iota_{i_w=1}\cdot\rho \tag{19}$$
$$\theta_{n,t} = \bar{m}\left(\frac{v_t}{u_t}\right)^{1-\mu} \tag{20}$$

Note that we can write the above value functions explicitly as a system of sub-problems: $P^{i_w=0}_{x<x_u^*}$, $P^{i_w=0}_{x\geq x_u^*}$, $P^{i_w=1}_{x<x_n^*}$, $P^{i_w=1}_{x\geq x_n^*}$, $P^{i_w=2}_{x<x_u^*}$, $P^{i_w=2}_{x\geq x_u^*}$, which correspond to the dynamic programing problem for i). previously self-employed with $x < x_u^*$; ii). previously self-employed with $x \geq x_u^*$; iii). previously employed with $x < x_n^*$; iv). previously employed with $x \geq x_n^*$; v). previously unemployed with $x < x_u^*$; vi). previously unemployed with $h \geq h_u^*$. To economize the space, we only provide the details for

$P^{i_w=0}_{x<x^*_u}$, and $P^{i_w=0}_{x\geq x^*_u}$ as follows.
$\left(P^{i_w=0}_{x<x^*_u}\right)$:

$$\mathbf{V}^1(h,z,x<x^*_u,a,i_w=0)=\max_{a'_n,a'_u}\left\{\begin{array}{c} p(\theta)\left(u(c_n)+\beta\mathbb{E}\mathbf{V}^0(h,z',x',a'_n,i'_w=1)\right) \\ +(1-p(\theta))\left(u(c_u)+\beta\mathbb{E}\mathbf{V}^0(h,z',x',a'_u,i'_w=0)\right)\end{array}\right\} \tag{21}$$

s.t.

$$c_n+a'_n\leq(1-\tilde{r}-\delta)a+zw(1-(1-\psi)\tau),\text{if matched with a firm} \tag{22}$$
$$c_u+a'_u\leq(1-\tilde{r}-\delta)a+b,\text{if unemployed} \tag{23}$$

$\left(P^{i_w=0}_{x\geq x^*_u}\right)$:

$$\mathbf{V}^1(h,z,x\geq x^*_u,a,i_w=0)=\max_{a'_e,k,n_h\geq 0}\left\{u(c_e)+\chi_{se}\mathbf{1}\{n_h=0\}+\chi_{emp}\mathbf{1}\{n_h>0\}+\beta\left[(1-\rho_e(h))\mathbb{E}\mathbf{V}^0(h\right.\right. \tag{24}$$

s.t.

$$c_e+a'_e\leq(1-\tilde{r}-\delta)a+\pi(k,n_h,x;\tilde{r},w,\tau) \tag{25}$$

where

$$\pi(k,n_h,x;\tilde{r},w,\tau)=xk^\alpha\left(\ell_e+\xi zn_h\right)^\gamma-w(1+\psi\tau)zn_h-rk-\frac{\kappa\rho zn_h}{q(\theta)}-f_{hire}\mathbf{1}\{n_h>0\}. \tag{26}$$

(d) Stop if $\max\left\{\left|\mathbf{V}^1-\mathbf{V}^0\right|,\left|f^1-f^0\right|\right\}\leq\epsilon$. Set $\mathbf{V}^*=\mathbf{V}^1$, and $f^*=f^1$. Otherwise, set $\mathbf{V}^0=\mathbf{V}^1$, $f^0=f^1$ and repeat from step (c).

3. Generate a large number of individuals, $N=100000$. For each agent j assign a vector of initial condition $\left(h^j_t,z^j_0,x^j_0,a^j_0,i^j_{w,0}\right)$, where $x^j_0\sim\Omega_x$, $z^j_0\in\Omega_z$, $i^j_{w,0}\in\Omega_w$.

4. Simulate the economy for T periods, where T is sufficiently large.

5. Calculate the following statistics from the simulated path $\left\{h_t^j, z_t^j, x_t^j, a_t^j, i_{w,t}^j, \mathcal{I}_{e,t}^j, n_t^j, k_t^j\right\}_{t=0}^{T}$.

$$LS^0 = \frac{\sum_{j=1}^{N}\left(\iota_{i_w=1}\cdot z^j\right)}{N} \tag{27}$$

$$KS^0 = \frac{\sum_{j=1}^{N}\left(\iota_{i_w=0}\cdot a^j + \iota_{i_w=1}\cdot a^j + \iota_{i_w=2}\cdot a^j - \iota_{i_w=0}\cdot k^j\right)}{\sum_{j=1}^{N} a^j} \tag{28}$$

$$u_t = \left(\int_h \iota_{i_w=0} + \int_h \iota_{i_w=2}\right)\cdot\int_0^{h_u^*} \tag{29}$$

$$v_t = \int_h \iota_{i_w=1}\cdot\rho \tag{30}$$

$$p(\theta)_t = \bar{m}\left(\frac{v_t}{u_t}\right)^{1-\mu} \tag{31}$$

$$G = \int \tau w z \tag{32}$$

and τ_y^1 that balances the government's budget.

6. Stop and set $(r^*, w^*, \theta^*) = \left(r^0, w^0, \theta^0\right)$, if $\max\left\{LS^0, KS^0, \left|\theta_n^1 - \theta_n^0\right|\right\} \leq \epsilon$. Otherwise, update aggregate variables (restart from step 2):

$$\begin{aligned} r^0 &= \chi r^0 + (1-\chi)r^1\left(KS^0\right) \\ w^0 &= \chi w^0 + (1-\chi)w^1\left(LS^0\right) \\ p(\theta)^0 &= \chi p(\theta)^0 + (1-\chi)p(\theta)^1 \\ \tau_y^0 &= \chi\tau_y^0 + (1-\chi)\tau_y^1 \end{aligned}$$

where $\chi \in (0,1)$ is the step for updating aggregate variables.

Appendix Search and Matching

I am not sure how much of this is needed for coding purposes but this is the standard search and matching model. The stock of employed workers n_t can be expressed as the current pool of workers who did not separate from firms, plus new hires

$$n_t = (1-\rho)n_{t-1} + m_{t-1}$$

The matching function is Cobb-Douglass where

$$m_t = \bar{m}v_t^{1-\mu}u_t^{\mu}$$

$\bar{m}$ is a match efficiency parameter , v is vacancies, u is unemployment, and match elasticity is $0 < \mu < 1$.The hiring rate can be defined for firms as total number of new hires to the number of vacancies

$$q(\theta_t) = \frac{m_t}{v_t} = \bar{m}\theta_t^{-\mu}$$

where $\theta_t = \frac{v_t}{u_t}$is aggregate labour market tightness. Similarly we can define this from the perspective of the worker in the job-finding rate.

$$p(\theta_t) = \frac{m_t}{u_t} = \bar{m}\theta_t^{1-\mu}$$

The Job Creation Condition determines the amount of new vacancies an employer will post. In our model, most workers are on rolling contracts, so firms only need to post new vacancies after some workers separate from these rolling contracts. Usually in search and matching models the wage is determined ex-post of matching by Nash bargaining between the employer and employee. However, in our span-of-control model the wage has already been determined ex ante, as the wage is an endogenous variable that determines occupational choice. The marginal benefit that the new employee would bring to the firm is expressed as A (which can be derived from the firms production function, $\gamma X k^{\alpha}(zn)^{\gamma-1}$). The JCC can be then expressed as the costs of replacing out workers on the LHS and the expected benefit on the R

$$\frac{\kappa}{q(\theta_t)} = A - w_t + \beta E_t\left[A - w_{t+1} + (1-\rho)\frac{\kappa}{q(\theta_{t+1})}\right]$$

where κ is the cost of posting a vacancy.

Matching Steady States

To find the steady state of the employment equation $n_t = (1-\rho)n_{t-1} + m_{t-1}$, we use the fact that $u = (1-n)$ and $m_t = \bar{m}v_t^{1-\mu}u_t^{\mu}$. Substituting and getting rid of time subscript gives $(1-u) = (1-\rho)(1-u) + \bar{m}v^{1-\mu}u^{\mu}$ which simplifies to $\bar{m}v_t^{1-\mu}u_t^{\mu}$ $=\rho(1-u)$. Making v the subject of the foruma gives us the Beveridge curve.

$$v = \left(\frac{\rho(1-u)}{\bar{m}u}\right)^{1-\mu} u$$

The steady state for the JCC can be found by using $\frac{\kappa}{q(\theta_t)} = A - w_t + \beta E_t\left[A - w_{t+1} + (1-\rho)\frac{\kappa}{q(\theta_{t+1})}\right]$ and $q(\theta_t) = \bar{m}\theta_t^{-\mu} = \bar{m}\left(\frac{v_t}{u_t}\right)^{-\mu}$. Substituting and getting rid of time subscripts yields

$$\frac{\kappa}{\bar{m}}\left(\frac{v}{u}\right)^{\mu} = A - w_t + \beta\left[A - w + (1-\rho)\frac{\kappa}{\bar{m}}\left(\frac{v}{u}\right)^{\mu}\right]$$

Simplifying yields the steady state of the JCC

$$1 - \beta(1-\rho)\frac{\kappa}{\bar{m}}\left(\frac{v}{u}\right)^{\mu} = (1+\beta)\left[A - w\right]$$

References

Buera, F. J. (2009). A dynamic model of entrepreneurship with borrowing constraints: theory and evidence. *Annals of Finance*, 5(3–4):443–464. 2

Fairlie, R. and Fossen, F. (2019). *Defining Opportunity versus Necessity Entrepreneurship: Two Components of Business Creation*. 2

Hurst, E. and Pugsley, B. W. (2011). What do small businesses do? *Brookings Papers on Economic Activity*, 2011(2):73–118. 2

Petrongolo, B. and Pissarides, C. A. (2001). Looking into the black box: A survey of the matching function. *Journal of Economic Literature*, 39(2):390–431. 2

Sedlacek, P. and Sterk, V. (2017). The growth potential of startups over the business cycle. *American Economic Review*, 107(10):3182–3210. 2